

# Theta Gainers Algo — Product Scope Document

**Audience:** the developer building the trading platform. **Purpose:** complete functional + UX specification with no code references. **Source of truth:** this doc supersedes any prior brief if there's a conflict. **Last updated:** 2026-04-30 **Owner:** Rohan Shah ( rohan@navingroup.in )

## How to read this document

- Sections **1–4** are the product overview, glossary and personas. Read once.
- Section **5 (Trade)** is the heart of the product — read carefully.
- Sections **6 (Dashboard)** and **7 (Reports)** are the other two priorities.
- Sections **8–14** cover supporting features.
- Section **15** lists every business rule and formula.
- Section **16** is the SEBI compliance checklist — non-negotiable.
- Section **17** is the acceptance test list.

When ASCII layouts appear, they show *relative position*, not pixel-precise design. The reference implementation (in this repo) is the visual spec — open `/trade` in the browser to see how it should feel.

## 1. Product overview

We sell **deep-OTM index options** (NIFTY, SENSEX) for premium decay, intraday and short-DTE. The platform must:

- Place orders into **multiple Indian brokers** from one screen (Axis Direct, Zerodha, Monarch, JM Financial; Paper for testing).
- Route orders across **multiple demat accounts** the trader has access to.
- Enforce SEBI compliance (rate cap, OTR, algo-ID tagging, freeze-qty iceberg slicing).
- Run a **Default Strategy** the trader can fire with one click — eliminates manual mistakes.
- Run a **Manual builder** for traders who want full control over strikes, lots, triggers and exits.
- Show a **live margin gauge** so the trader never overcommits capital.
- Hash-chain every state change for audit (pre-trade, fills, exits, kill events).
- Send multi-channel alerts (WhatsApp / Telegram / Email / SMS) on key events.

## 2. Glossary

| Term               | Meaning                                                            |
|--------------------|--------------------------------------------------------------------|
| <b>Underlying</b>  | NIFTY or SENSEX index.                                             |
| <b>Spot</b>        | Current value of the underlying (e.g. NIFTY 24,812).               |
| <b>Strike</b>      | The strike price of an option contract (e.g. 25,600 CE).           |
| <b>CE / PE</b>     | Call / Put option.                                                 |
| <b>Lot size</b>    | Exchange-defined units per lot. NIFTY = 65, SENSEX = 20 (current). |
| <b>Strike grid</b> | Allowed strike interval. NIFTY = 50, SENSEX = 100.                 |

|                   |                                                                           |
|-------------------|---------------------------------------------------------------------------|
| <b>Freeze qty</b> | Max units per single order. NIFTY = 1,800, SENSEX = 1,000.                |
| <b>OTM</b>        | Out-of-the-money. CE is OTM if strike > spot. PE is OTM if strike < spot. |
| <b>Premium</b>    | The price (₹) you receive for selling an option.                          |
| <b>Margin</b>     | Capital blocked by the broker to support a short position.                |
| <b>SOR</b>        | Smart Order Routing — splitting orders across brokers/demats.             |
| <b>OTR</b>        | Order-to-Trade Ratio — SEBI watches this, penalty above ~250.             |
| <b>Algo-ID</b>    | A code SEBI requires every algo order to carry.                           |
| <b>Iceberg</b>    | Splitting a single order > freeze qty into multiple sub-orders.           |
| <b>Strangle</b>   | Selling 1 OTM CE + 1 OTM PE simultaneously.                               |

## 3. User personas & permissions

Three trader profiles. Same login screen; UI changes based on `permissions` returned by `/admin/me/permissions`.

### 3.1 Default-only Trader (most restrictive)

- Sees: Free margin strip, Broker/Demat picker, **Default Strategy CTA only**, Cancel.
- Cannot see: Manual leg builder, custom triggers, custom exits, Save Draft, Start Monitor, Execute Now.
- Use case: junior team member; risk-controlled; one-click trader.

### 3.2 Full Trader (default for senior team)

- Sees everything in the Trade page: Default + Manual builder + all configurability.
- Can set custom triggers, exit rules, multi-broker SOR, allocations.
- Approval required if any leg has lots  $\geq 5$  (configurable).

### 3.3 Admin

- Same as Full Trader **plus** Settings (brokers, risk, execution, users), Admin (audit browser, user permissions).
- Can grant / revoke `default_only` flag on any user.
- Can act as second approver for  $\geq 5$ -lot orders.
- Can access kill-switch on any running strategy.

## 4. Top-level navigation

Sidebar layout (left rail), persistent across all pages:

|                |                                  |
|----------------|----------------------------------|
| Theta Gainers  |                                  |
| v1.0.0 [PAPER] |                                  |
| ■ Dashboard    | ← P&L, active strategies summary |

|                 |                                   |
|-----------------|-----------------------------------|
| ⊗ Trade         | ← THE main page                   |
| + Templates     | ← saved strategy presets          |
| + Analytics     | ← Deep OTM strike recommendations |
| ■ Reports       | ← live + paper, by strategy/demat |
| ⚙ Settings      | ← brokers, risk, execution, users |
| 🛡 Admin (admin) | ← audit browser, permissions      |
| [theme toggle]  |                                   |
| Signed in as    |                                   |
| rohan · ADMIN   |                                   |
| [ Logout ]      |                                   |

Top bar (above content):

- Live market strip: NIFTY · SENSEX · BANKNIFTY · VIX with % change, MaxPain, FII/DII, PCR
- Auto-scrolling news ticker (NSE / SEBI / Reuters / Mint / Quantsapp)
- Header: connected broker chip ( PAPER · PAPER-001 ), SEBI rate counter ( SEBI 0/8 ), OTR ( OTR 2.1 )

## 5. Trade page (the heart of the product)

Path: /trade (also /strategy/new aliased to same page).

Vertical stack of sections, top to bottom:

[ Page header: "Trade" + Save as Template / Load Template ]

Free margin strip

Free margin: ₹6.30L (usable for new strategy)

Total ₹10.00L · Used (active) ₹3.25L · Blocked (pending) ₹0.45L

[  63% free ]

New strategies are sized only against free margin.

Broker & Demat (global – applies to every order on this page)

☐ Multi-broker SOR    ☐ Multi-demat

Broker: [Zerodha (Kite Connect) ▼]    Demat: [ZD12345 ▼]

Margin allocation (when multi-\* on):

Deploy [   ] Cr · keep [5]% or [500000] ₹ free per demat

→ Pulls weighted: ZD12345 ₹2.1L, ZD67890 ₹3.9L, MN98765 ₹0.5L

Default Strategy CTA (recommended path)

 Default Strategy · Deep OTM Strangle

Spot 24,812   CE strike 25,600   PE strike 24,050   Distance ≥ 2.5% OTM

Lot 65u   Margin/lot ~₹105K   Target ₹5K/Cr   Routes via ZERODHA·ZD12345

Strikes round further from spot on the 50-pt grid.

|                  |
|------------------|
| Symbol [NIFTY]   |
| Lots [1]         |
| [Load Strategy]  |
| [Load + Execute] |

(Manual builder – hidden if user is Default-only)

Strategy basics

Strategy Name [Short Strangle · NIFTY 17-Apr]

Underlying: ☒ NIFTY ☐ SENSEX

Portfolio: [Default Portfolio ▼]

Strike Selection mode: [Manual] [Automatic (Rule Builder)]

(Manual selected)

Legs (2)

+ Add Leg

| Σ                                   | B/S   | Expiry   | Strike         | Type  | Lots | Order | LTP   | Trade Price | Actions                                                                                                                                                                                                                                                           |
|-------------------------------------|-------|----------|----------------|-------|------|-------|-------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | [B S] | 17 Apr ▼ | [–][25,600][+] | CE/PE | [1]  | LIMIT | 18.40 | [18.40]     |    |
| <input checked="" type="checkbox"/> | [B S] | 17 Apr ▼ | [–][24,050][+] | CE/PE | [1]  | LIMIT | 22.85 | [22.85]     |    |

[Adjust: Shift / Width / Hedge]
Multiplier [1▼]

Footer: Legs in Σ 2/2 · Net Qty 130u · Credit ₹5,200 · iceberg n/a

Reset Prices

(Click  on a leg to expand: shows Bid/Ask/Spread/OI/Volume + OHLC + intraday snapshots)

Entry Time Window

☒ Restrict by time

Entry from [09:30] Entry to [10:30]

Presets: [Open] [Morn] [Mid] [Aft] [All]

Premium Trigger

[ Combined Σ ]
[ Per ₹1Cr ]
[ Per-leg ]
[ Enter now ]

(Per ₹1Cr selected, for example)

Combined premium per ₹1Cr margin ≥ ₹ [5000]

Live ratio: ₹4,650/Cr · Threshold ₹5,000/Cr · [Waiting]

Formula: (Σ leg credit × 65) ÷ margin × ₹1Cr.

Exit Rules & Kill Switches

Stop Loss [3000] Target [2000] Square-off [15:15] MTM-DD [40]%

- ☐ Trailing SL (when on: Activate after profit ₹ + Step ₹)
- ☐ Lock-in profits (when on: when profit ≥ ₹X, move SL to BE)
- ☐ Exit when spot approaches strike ← NEW
  - Distance type: [Points] [%]
  - Threshold: [150] pts (or [0.5] %)
  - When triggered: [Exit that leg only] [Exit both legs]
- Advanced – Dead-man switch (off by default; HFT only)

Live Margin Gauge

This strategy will use ₹2.10L of ₹6.30L free

[  33% of free margin ] ₹4.20L will remain

(Red bar + warning if exceeds free margin; submit buttons disable)

Sticky action bar (bottom)

2 legs · 130u · Credit ₹5,200 · Waiting for combined trigger

[Cancel] [Save Draft] [Start (Monitor)]

### 5.1 Free margin strip (top)

**REQ-T01.** Always shows: Free / Total / Used by active / Blocked by pending. Live, refreshes every 15 s. **REQ-T02.** Bar shows used (solid) + blocked (translucent) overlay; the "free %" label sits above the bar. **REQ-T03.** Footer note: "New strategies are sized only against free margin. Pre-trade RMS rejects orders that exceed it."

### 5.2 Broker & Demat (global)

**REQ-T10.** Two toggles, top-right of the section:

- **Multi-broker SOR** (off by default)
- **Multi-demat** (off by default; auto-enables when Multi-broker is on)

**REQ-T11.** When **both off**: single broker dropdown + single demat dropdown.

**REQ-T12.** When **multi-demat on (single broker)**: demats render as a checkbox list with cap badges. SOR splits across selected demats by free margin.

**REQ-T13.** When **multi-broker on**: brokers render as expandable rows; checking a broker reveals its demats inline. Selected broker × demat combinations all participate in SOR.

**REQ-T14.** Selection routes **every** order from this page — Default, Manual, Execute Now, trigger-based starts.

**REQ-T15.** Margin allocation row appears only when multi-\* is on. Single sentence shape: "Deploy [X] Cr · keep [Y]% or [Z]₹ free per demat" Below: live preview line listing allocated ₹ per demat.

**REQ-T16.** Cushion math: per demat, reserve =  $\max(\text{balance} \times Y\%, Z₹)$  . Deployable = balance – reserve. Total cap =  $\min(\text{deployable across all selected}, X \text{ Cr})$  . Allocation = weighted by deployable.

**REQ-T17.** Bottom note: "This selection routes every order placed from this page."

### 5.3 Default Strategy CTA (recommended path)

**REQ-T20.** Card with subtle accent tint (not a loud banner). Sub-title: *"Deep OTM Strangle"*.

**REQ-T21.** Left column shows 8-cell key-value grid: Spot · CE strike · PE strike · Distance · Lot size · Margin/lot · Target · Routes via.

**REQ-T22.** Strike rule (text + behaviour):

- CE rounds **up** to next grid (further from spot above)
- PE rounds **down** to next grid (further from spot below)
- Never closer than `DEFAULT_DISTANCE_PCT` (default 2.5%).
- Concrete example shown in HANDOFF / SCOPE: 3% = 24,535 → 24,600 CE; 21,230 → 21,200 PE.

**REQ-T23.** Right column has 4 controls:

1. **Symbol** dropdown: NIFTY (lot 65, grid 50) / SENSEX (lot 20, grid 100). Switching updates spot, lot, grid, margin.
2. **Lots** dropdown: 1 / 2 / 3 / 5 / 10 / 15 / 20.
3. Primary button: **Load Strategy** (loads pre-fill, no orders).
4. Danger button: **Load + Execute** (loads + immediately fires Execute Now confirm).

**REQ-T24.** Both buttons open a confirm modal with full details (see §5.10).

**REQ-T25.** When user is **default-only**, this is the only action area visible on the page (manual builder hidden, sticky bar shows Cancel only).

## 5.4 Strategy basics

**REQ-T30.** Editable fields: Strategy Name (free text), Underlying (NIFTY / SENSEX radio), Portfolio (assigned per user).

## 5.5 Strike Selection mode

**REQ-T40.** Two-button toggle: **Manual** (default) or **Automatic (Rule Builder)**.

**REQ-T41.** Manual mode: legs table directly editable.

**REQ-T42.** Automatic mode reveals a rule builder above the legs table:

- **Primary criteria** card (the focal point):
  - Distance metric: **% away** / **Points** / **Delta** (radio)
  - Same for CE & PE or Independent CE / PE (toggle)
  - Single big input (or two side-by-side inputs when Independent)
- **Advanced filters** collapsible (off by default): rule tree of OI, spread, volume, IV rank, regime, time window etc. with ALL-of / ANY-of / NOT combinators. CE & PE rules can mirror or be independent.
- **Live Preview** panel shows candidate strikes with PASS/FAIL badges, price, OI, "Use" button to load into legs.

**REQ-T43.** Auto-trade fires only when both **strike rule** AND **premium trigger** are satisfied.

## 5.6 Legs table

**REQ-T50.** Table columns (centred where relevant):  $\Sigma$  · B/S · Expiry · Strike+ · Type · Lots · Order · LTP · Trade Price · Actions.

**REQ-T51.** Column behaviours:

- $\Sigma$  checkbox: include this leg in the combined-premium trigger sum.

- **B/S**: 2-button toggle (Buy/Sell) with red SELL / green BUY tint.
- **Expiry**: dropdown of available weeklies + monthly.
- **Strike**: – / value / + with grid increment from underlying (50 NIFTY / 100 SENSEX).
- **Type**: 2-button toggle CE / PE with red CE / blue PE tint.
- **Lots**: integer dropdown 1–30.
- **Order**: LIMIT / LIMIT+buf / MARKET (MARKET disabled with tooltip — SEBI safety).
- **LTP**: read-only, live price.
- **Trade Price**: editable LIMIT price; **also serves as the per-leg premium threshold** when trigger mode is "Per-leg". Disabled in Combined  $\Sigma$  / Per ₹1Cr modes (tooltip explains why). Accent border in Per-leg mode.
- **Actions**:  expand ·  duplicate ·  remove.

**REQ-T52.** Add Leg button disabled in Combined  $\Sigma$  / Per ₹1Cr modes when there are already 2 legs (lock to CE+PE strangle); tooltip explains.

**REQ-T53.** When trigger mode switches to Combined  $\Sigma$  or Per ₹1Cr, extra legs auto-trim to 2 with a toast.

**REQ-T54.** Expanded view (click ) shows three sub-rows:

1. **Live Quote**: Bid / Ask / Bid Qty / Ask Qty / Spread % / OI / Volume.
2. **Today's Range**: Open / High / Low / Prev Close / % Chg / IV / Delta / Theta.
3. **Intraday Snapshots (IST)**: prices at 09:20 / 09:45 / 10:30 / 11:00 / 12:00.
4. **Per-leg threshold input** (only in Per-leg mode): bid  $\geq$  ₹X.

**REQ-T55.** Footer summary: Legs in  $\Sigma$  count / Net Qty / Credit-Debit / iceberg chip if needed / **Reset Prices** button.

## 5.7 Entry Time Window

**REQ-T60.** Toggle "Restrict by time" (default ON, value 09:30–10:30).

**REQ-T61.** From / To time inputs (24h IST).

**REQ-T62.** Quick presets: Open (09:15–09:30), Morn (09:30–10:30), Mid (11:00–13:00), Aft (13:30–14:30), All (09:15–15:15).

**REQ-T63.** When off, Premium Trigger description picks up the change: *"No time restriction — entries can fire anytime trigger conditions are met."*

## 5.8 Premium Trigger

Four modes, segmented toggle top-right:

### 5.8.1 Combined $\Sigma$

**REQ-T70.** Threshold input: *Combined  $\geq$  ₹X (sum of  $\Sigma$ -marked legs).*

**REQ-T71.** Live displays:

- Live sum of N legs: ₹A.BB
- Threshold: ₹X
- Status chip: **MET** (green) / **Waiting** (yellow), with pulsing icon when MET.

**REQ-T72.** Formula:  $\Sigma(\text{SELL leg bid}) - \Sigma(\text{BUY leg ask})$  for  $\Sigma$ -marked legs.

**REQ-T73.** Locks to 2 legs (CE + PE strangle).

### 5.8.2 Per ₹1Cr

**REQ-T80.** Threshold input: *Combined premium per ₹1Cr margin  $\geq$  ₹X* (default 5,000).

**REQ-T81.** Live ratio displayed as `₹A,BCD / Cr`. Status chip same as Combined.

**REQ-T82.** Formula: `( $\Sigma$  leg credit  $\times$  lot_size) / margin_required  $\times$  ₹1Cr`.

**REQ-T83.** Locks to 2 legs (same as Combined).

**REQ-T84.** This is the trader's primary discipline: yield-on-margin. Mirrors the Default Strategy's ₹5K/Cr target so manual mode can replicate.

### 5.8.3 Per-leg

**REQ-T90.** No duplicate threshold inputs in this section. Instead, a read-only summary listing each leg's live threshold (pulled from the leg's **Trade Price** in the Legs table — single source of truth).

**REQ-T91.** Execution sub-toggle:

- **Both legs together** (default): both CE and PE must be at threshold simultaneously to fire the strategy.
- **Each leg independent**: each leg fires on its own when its threshold is met; the other leg can wait.

**REQ-T92.** "Examples" collapsible panel maps the trader's three concrete cases to the right combination of strike + trigger settings.

### 5.8.4 Enter now

**REQ-T95.** Bypass trigger; place LIMIT immediately (subject to RMS + margin check).

#### Trigger / time / strike interaction

**REQ-T96.** Trigger evaluation only runs **inside the entry time window** (when restricted) AND when **strike rule passes** (auto mode). All three must agree before an order fires.

## 5.9 Exit Rules & Kill Switches

**REQ-T100.** Default-visible row (always editable):

- Stop Loss (₹)
- Target (₹)
- Square-off time (IST) — defaults to 15:15
- MTM Drawdown kill (% from peak) — default 40%

**REQ-T101.** Trailing SL toggle row:

- When on: "Activate after profit ₹X" + "Step ₹Y".
- Move SL up by step every X profit reached.

**REQ-T102.** Lock-in profits toggle row:

- When on: "When profit  $\geq$  ₹X, move SL to breakeven".

**REQ-T103.** Exit when spot approaches strike toggle row (NEW — defensive exit):

- **Distance type**: Points / %
- **Threshold**: e.g. 150 pts or 0.5 %
- **When triggered**: *Exit that leg only / Exit both legs*

- Helper text walks through example: *"Sold CE 25,600, proximity 150 pts (leg-only). If spot rises to 25,450, this leg auto-exits."*
- Math: points  $\rightarrow |spot - strike| \leq threshold$  . Percent  $\rightarrow |spot - strike|/strike \times 100 \leq threshold$  .

**REQ-T104.** Advanced collapsible: **Dead-man switch** (off by default). Helper text says *"Not recommended for Deep OTM strangle — your SL, MTM-DD kill, and square-off time already cover unattended cases without false-trigger risk from network blips or laptop sleep."*

## 5.10 Confirm modals

Every action that places orders or persists state shows a confirm modal. Each modal includes:

- Underlying, Spot
- Broker · Demat (from global selection)
- Strategy name, exact strikes, lots, units per leg
- Trigger config summary
- Estimated credit/debit, margin required, free margin
- Exit rules summary
- Two-person approval warning when lots  $\geq 5$

### Modal types

**REQ-T110. Load Default Strategy** confirm: title *"Load Default Strategy?"* (info tone). Body lists pre-fill plan. Confirm label: *"Load configuration"*.

**REQ-T111. Load + Execute** confirm: title *"Default Strategy — Execute Now?"* (danger tone). Confirm requires typing **EXECUTE** to enable submit.

**REQ-T112. Start (Monitor)** confirm: title *"Start Strategy?"* (info tone). Body explains engine will poll quotes every 2 s and submit when trigger met.

**REQ-T113. Execute Now** (manual, from sticky bar) confirm: title *"Execute Now — place orders immediately?"* (danger tone). Body says *"This bypasses the trigger and places all leg orders right now at LIMIT prices. Real money."* Type **EXECUTE** to enable.

**REQ-T114. Save Draft** confirm: small modal, info tone, confirms name. Strategy persists in DRAFT state.

**REQ-T115. Cancel** confirm: warn tone, *"Discard changes?"*.

## 5.11 Sticky action bar

**REQ-T120.** Always visible at bottom of viewport.

**REQ-T121.** Left side summary: legs · units · credit/debit · trigger status text.

**REQ-T122.** Right side actions:

- Cancel (always)
- Save Draft (hidden in default-only mode)
- Start (Monitor) **OR** Execute Now (depends on trigger mode; hidden in default-only)
- Buttons disable when over-margin with tooltip showing required vs free.

## 5.12 Live margin gauge

**REQ-T130.** Sits between exit rules and sticky bar. Compares this strategy's required margin to free margin.

**REQ-T131.** Coloured progress bar:

- Green: < 80% of free margin used.
- Amber: 80–100%.
- Red: > 100% (over budget). Submit disabled.

**REQ-T132.** Right-side label: "₹X.XL will remain" (positive cushion) or "OVER by ₹X.XL".

**REQ-T133.** Hidden in default-only mode (those traders use Default CTA which has its own margin display).

### 5.13 Templates / Save / Load

**REQ-T140.** Top-right of page: **Save as Template** + **Load Template...** dropdown.

**REQ-T141.** Built-in templates (provided): Short Strangle, Iron Condor, Bull Put Spread, Calendar Spread.

**REQ-T142.** Saved templates listed in `/templates` page; loadable from any time.

## 6. Dashboard

Path: `/`

The home page. Read-only summary; no order placement happens here.

Header strip

Welcome back, Rohan · 30 Apr 2026 · NIFTY 24,812 (+0.32%)

KPI cards (4-up grid)

Today P&L  
+₹14,250  
▲ 1.42%

Open Strategies  
3 active

Margin Free  
₹6.30L

Win rate (30)  
68% · 17/25 trades

Active Strategies

[ View All → ]

| ID   | Name                 | State      | P&L    | Trigger | Actions |
|------|----------------------|------------|--------|---------|---------|
| S-42 | Short Strangle NIFTY | LIVE       | +₹2.4K | filled  | [Exit]  |
| S-43 | Default · SENSEX     | MONITORING | n/a    | waiting | [Exit]  |
| S-44 | Iron Condor NIFTY    | EXITING    | -₹0.8K | partial | [-]     |

Today's P&L curve

[intraday line chart, 1-min ticks, with entry/exit markers]

Peak +₹18,200 · Trough +₹2,100 · MTM-DD 12% (well under kill)

Quick stats (3-up)

Risk usage 31% · OTR 2.1 · SEBI orders today 18/640

Recent Activity

[ View All → ]

10:32 Strategy S-43 entered MONITORING

10:31 Trade fired: SELL 1× 25,600 CE @ ₹18.40 · ZD12345

|       |                                               |
|-------|-----------------------------------------------|
| 10:30 | Strategy S-43 created (Default · SENSEX)      |
| 10:18 | S-42 trailing SL moved to ₹2,500 (was ₹3,000) |

Notifications inbox (last 5)

|          |                                                |
|----------|------------------------------------------------|
| [📱 ✉️ 💬] | Today 10:31 – Strategy S-43 entered MONITORING |
| ...      |                                                |

## Dashboard requirements

**REQ-D01. Today P&L card:** signed ₹ value + intraday % return + trend arrow. Refreshes every 5 s.

**REQ-D02. Open Strategies card:** count of strategies in {DRAFT, MONITORING, LIVE, EXITING}. Click → /reports filtered to active.

**REQ-D03. Margin Free card:** same number as the Trade page strip. Click → /trade.

**REQ-D04. Win rate (30 days) card:** % of profitable closed strategies in last 30 trading days + raw counts.

**REQ-D05. Active Strategies table:**

- Columns: ID · Name · State · P&L (signed, coloured) · Trigger status · Actions (Exit / Kill).
- Sortable by any column. Default sort: P&L descending (winners on top).
- Each row links to /strategy/:id for live monitor.
- **Kill** action requires confirmation typing the strategy ID.

**REQ-D06. P&L curve:**

- Line chart, intraday minute-level data.
- Entry/exit markers on the curve.
- Annotations: Peak, Trough, current MTM drawdown %.
- Below the chart: today's all-strategies aggregate, not per-strategy.

**REQ-D07. Quick stats strip:**

- **Risk usage** =  $\text{used\_margin} / \text{total\_margin} \times 100$ .
- **OTR** = orders / trades for today (live counter).
- **SEBI orders today** = orders sent today / 640 (8/sec × ~80 sec/min × 80 min budget).

**REQ-D08. Recent activity feed:** chronological, top 10 events, links to source (strategy detail, audit entry).

**REQ-D09. Notifications inbox:** last 5 notifications across all channels with channel icons. Click → full inbox view.

**REQ-D10. Auto-refresh:** KPIs every 5 s; tables every 15 s; chart streams via WebSocket.

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## 7. Reports

Path: /reports/\*

Reporting is **algo-only** in this app (paper + live). The separate F&O reporting tool is a different product.

### 7.1 Tab structure

[ Trades ] [ Strategies ] [ By Demat ] [ By Portfolio ] [ Tax ]

## 7.2 Trades tab

|                                                                    |       |         |       |        |      |      |      |       |       |      |
|--------------------------------------------------------------------|-------|---------|-------|--------|------|------|------|-------|-------|------|
| Filters                                                            |       |         |       |        |      |      |      |       |       |      |
| Date: [01 Apr] – [30 Apr]    Mode: [All] [Live] [Paper]            |       |         |       |        |      |      |      |       |       |      |
| Underlying: [All] [NIFTY] [SENSEX]                                 |       |         |       |        |      |      |      |       |       |      |
| Strategy: [All ▼]   Demat: [All ▼]   Direction: [All] [BUY] [SELL] |       |         |       |        |      |      |      |       |       |      |
| [Apply]   [Reset]   [↓ Export CSV]   [↓ Export PDF]                |       |         |       |        |      |      |      |       |       |      |
| Summary strip                                                      |       |         |       |        |      |      |      |       |       |      |
| Trades 142 · Net P&L +₹38,400 · Brokerage -₹2,140 · Net +₹36,260   |       |         |       |        |      |      |      |       |       |      |
| Live 84 · Paper 58 · Win rate 64% · Avg trade +₹270                |       |         |       |        |      |      |      |       |       |      |
| Trades table                                                       |       |         |       |        |      |      |      |       |       |      |
| Ts                                                                 | Strat | Demat   | Sym   | Strike | Type | Side | Lots | Px    | P&L   | Mode |
| 10:32:14                                                           | S-42  | ZD12345 | NIFTY | 25,600 | CE   | S    | 1    | 18.40 | +₹820 | LIVE |
| ...                                                                |       |         |       |        |      |      |      |       |       |      |
| (paginated, 50/page; each row links to audit entry)                |       |         |       |        |      |      |      |       |       |      |

**REQ-R10.** Filters: date range, mode, underlying, strategy, demat, direction.

**REQ-R11.** Live + Paper **clearly distinguished** — same table, **Mode** column always visible, paper rows muted (low contrast).

**REQ-R12.** Summary strip shows: Trade count, Gross P&L, Brokerage estimate, Net P&L, Live/Paper split, Win rate, Avg trade.

**REQ-R13.** Export: CSV (raw rows) and PDF (formatted summary + table).

**REQ-R14.** Each trade row links to its audit-log entry (full hash chain visible).

## 7.3 Strategies tab

|                  |              |        |       |        |         |          |      |
|------------------|--------------|--------|-------|--------|---------|----------|------|
| Strategies table |              |        |       |        |         |          |      |
| ID               | Name         | State  | Open  | Closed | Net P&L | Avg Hold | Mode |
| S-42             | Short Strang | CLOSED | 10:30 | 14:55  | +₹2,400 | 4h 25m   | LIVE |
| ...              |              |        |       |        |         |          |      |

**REQ-R20.** One row per strategy run.

**REQ-R21.** Click strategy → drill-down: leg-by-leg fills, all events on timeline, exit reason, slippage, brokerage breakdown.

**REQ-R22.** Group-by toggles: by day / by week / by underlying / by demat / by mode.

## 7.4 By Demat tab

**REQ-R30.** Aggregate P&L per demat, per broker, with charge breakdown (brokerage, STT, exchange, GST, SEBI fee, stamp).

**REQ-R31.** Visualised as horizontal bar chart of net P&L per demat for the selected period.

**REQ-R32.** Reconciliation column showing app-computed P&L vs broker-reported P&L; diff highlighted.

### 7.5 By Portfolio tab

**REQ-R40.** When portfolios are set up in Settings → Users, group strategies by portfolio.

**REQ-R41.** Same metrics as Strategies tab but rolled up.

### 7.6 Tax tab (basic)

**REQ-R50.** Computes sum of: realised gains, brokerage, STT, exchange charges, GST. **Disclaimer:**  
*"Indicative only — confirm with your CA. Speculative income (intraday) and STCG treatment may apply."*

### 7.7 General report rules

**REQ-R60.** All reports have a date-range filter, default = today.

**REQ-R61.** All numeric columns sortable.

**REQ-R62.** Export to CSV must use the user's filter selection — no surprises.

**REQ-R63.** No real-time refresh on /reports/\* — load on filter apply.

**REQ-R64.** Hide /reports for default-only traders (admin permission).

---

## 8. Templates

Path: /templates

**REQ-X01.** List view of saved strategy presets with name, kind, leg count, last used date.

**REQ-X02.** Built-in presets always at top, user-saved below.

**REQ-X03.** Click → loads into Trade page with all settings pre-filled.

**REQ-X04.** Edit / Delete / Duplicate actions per saved preset.

---

## 9. Analytics (Deep OTM)

Path: /analytics

**REQ-Y01.** Strike recommendation board — for the chosen underlying + expiry, show top strikes per tier:

- Tier 1: 0.05–0.10 delta (deepest OTM)
- Tier 2: 0.10–0.15
- Tier 3: 0.15–0.20
- Tier 4: 0.20+

**REQ-Y02.** Per recommendation: strike, premium, OI, cushion ratio (= cushion ÷ expected move), POP estimate, score.

**REQ-Y03.** "Use" button on each recommendation loads it into the Trade page.

**REQ-Y04.** Refresh from live chain.

---

## 10. Settings

Path: `/settings/*`

### 10.1 Brokers & Demats

**REQ-S01.** List broker sessions per user. Each row: broker · last connected · session expiry · health.

**REQ-S02.** Connect / Reconnect / Disconnect actions. OAuth flow for Zerodha / Axis / Monarch / JM.

**REQ-S03.** Demat assignments per user: which demats can this user trade through.

### 10.2 Risk & Limits

**REQ-S10. Configurable kill switches** with three states each (Required / Optional / Disabled — admin-controlled):

- SL kill
- Target kill
- Time kill (square-off)
- MTM-DD kill
- Trailing SL
- Lock-in profits
- Spot proximity exit
- Dead-man switch
- OTR kill
- Reconcile failure kill
- Circuit breaker

**REQ-S11.** Per-user position limits: max lots per strategy, max strategies open, max margin %.

**REQ-S12.** Two-person approval threshold (default  $\geq 5$  lots).

### 10.3 Execution (OMS)

**REQ-S20.** Iceberg config: jitter (default 100 ms), max retries, peg behaviour.

**REQ-S21.** SEBI rate cap: orders/sec/user (default 8), orders/sec/global (default 20).

**REQ-S22.** Order kind defaults: LIMIT vs LIMIT+buf (with buffer %).

### 10.4 Users

**REQ-S30.** Admin manages users: create, edit, role (Admin / Trader / Risk Officer), active status, IP allowlist.

**REQ-S31.** Per-user permissions panel:

- `default_only` flag
- `can_execute_now` flag
- `can_use_multi_broker` flag
- `max_lots_without_approval` (int)

**REQ-S32.** Demat assignments per user (which demats they can route through).

**REQ-S33.** Reset password / Reset 2FA actions (require admin).

## 10.5 Notifications

**REQ-S40.** Per-user channels: WhatsApp / Telegram / Email / SMS / Voice.

**REQ-S41.** Per-event severity routing — choose which channels fire on each event class:

- **Critical:** all kill events, broker outage, RMS violation.
- **Warning:** partial fills, slippage > X%, MTM-DD approaching limit.
- **Info:** trigger met, fills, exits.

**REQ-S42.** Test-send button per channel.

## 10.6 Profile

**REQ-S50.** User can: change password, manage 2FA, view session history, manage API tokens.

---

# 11. Admin

Path: `/admin/*`

## 11.1 Audit Browser

**REQ-A01.** List audit entries with filters: date, user, event type, strategy, severity.

**REQ-A02.** Each entry shows: timestamp, user, action, before/after state diff, hash, prev\_hash.

**REQ-A03.** **Verify chain** button: walks the chain and reports first break (should never break).

**REQ-A04.** Export audit log range to CSV / JSON for SEBI inspection.

## 11.2 User Permissions

**REQ-A10.** Per-user grid showing all permission flags with toggles.

**REQ-A11.** Bulk actions: grant default-only to multiple users at once.

## 11.3 EOD Reconciliation

**REQ-A20.** Daily 18:00 IST job reconciles app trades vs broker contract notes; surfaces diffs > ₹10.

## 11.4 Live Operations

**REQ-A30.** **Kill all running strategies** — emergency button, requires 2nd admin approval.

**REQ-A31.** **Halt new orders** — keeps running strategies intact, prevents new submissions.

---

# 12. Notifications & alerting

**REQ-N01.** All channel sends go through a queue with retry (3 attempts, exponential backoff 5s/30s/2m).

**REQ-N02.** Templates per event in Settings → Notifications.

**REQ-N03.** WhatsApp via Twilio Business API; Telegram via Bot API; Email via SMTP/SendGrid; SMS via MSG91; Voice via Exotel.

**REQ-N04.** Every notification creates an audit entry ( `channel` , `template` , `payload_hash` , `delivered_at` , `attempts` ).

---

## 13. Authentication & onboarding

### 13.1 Login

**REQ-AU01.** Username + password. TOTP 2FA optional but recommended.

**REQ-AU02.** Exponential lockout: 5 fails → 1 min lock; 10 fails → 5 min; 20 fails → admin unlock.

**REQ-AU03.** IP allowlist enforced if user has one configured (set in Settings → Users).

**REQ-AU04.** Session cookie + JWT pair. Refresh token rotation.

### 13.2 First-time onboarding

**REQ-AU10.** After login, if user has no broker session → forced onto **Connect Broker** page.

**REQ-AU11.** After broker connect → redirect to /trade.

**REQ-AU12.** TOTP setup wizard on first login if 2FA not configured.

---

## 14. Mobile / PWA

**REQ-M01.** App must be installable as a PWA with manifest + service worker.

**REQ-M02.** Mobile breakpoint shows simplified Trade page: Default Strategy CTA only by default. Manual builder collapses behind an "Advanced" tab.

**REQ-M03.** Offline fallback page: *"Trading requires a live connection. Reconnect and try again. Already-placed orders continue on the server — nothing is lost."*

**REQ-M04.** Push notifications via Web Push API for critical events (browsers that support it).

---

## 15. Business rules & formulas

Centralised so there's no drift between UI and engine.

### 15.1 Strike rounding (REQ-T22, REQ-T42)

Given:

|      |                                      |
|------|--------------------------------------|
| spot | – current underlying price           |
| pct  | – distance % requested (e.g. 2.5)    |
| grid | – strike grid (50 NIFTY, 100 SENSEX) |

Raw CE strike =  $\text{spot} \times (1 + \text{pct}/100)$

Raw PE strike =  $\text{spot} \times (1 - \text{pct}/100)$

```
CE strike = ceil(raw_ce / grid) × grid    # round UP (further from spot)
PE strike = floor(raw_pe / grid) × grid    # round DOWN (further from spot)
```

The chosen strike is **never closer than** `pct` **away** from spot. If the raw value is exactly on a grid line, it stays.

## 15.2 Lot size + units

```
Units per leg = lots × lot_size
NIFTY lot_size = 65    (current)
SENSEX lot_size = 20    (current)
```

Lot size comes from exchange master; the figures above are seeds.

## 15.3 Iceberg slicing (REQ-Compliance)

```
freeze_qty    = 1800 (NIFTY) / 1000 (SENSEX)
slices_for_leg(units) = ceil(units / freeze_qty)
```

Slices are dispatched with **100 ms jitter** between sub-orders to avoid burst rate-limit hits.

## 15.4 Combined $\Sigma$ trigger

```
combined_live =  $\Sigma$ (SELL leg bid) -  $\Sigma$ (BUY leg ask)    # only  $\Sigma$ -marked legs
fires when: combined_live  $\geq$  threshold
```

## 15.5 Per ₹1Cr trigger

```
combined_credit_rupees =  $\Sigma$ (SELL leg bid - BUY leg ask) × lot_size × lots    #  $\Sigma$ -marked
ratio_per_cr           = combined_credit_rupees / margin_required × 1_00_00_000
fires when: ratio_per_cr  $\geq$  threshold
```

## 15.6 Per-leg trigger

```
For each leg with side=SELL:
    fires when: leg.bid  $\geq$  leg.trade_price
```

```
linked    : strategy fires only when ALL eligible legs are firing simultaneously
independent: each leg fires on its own; other legs can wait
```

## 15.7 Spot-approaches-strike exit (REQ-T103)

```
For each open leg:
    gap = abs(spot - leg.strike)
    if mode == "points":    fires when gap  $\leq$  threshold
    if mode == "percent":   fires when gap / leg.strike × 100  $\leq$  threshold
```

```
scope=leg : exit just this leg
scope=both : exit both legs of the strangle
```

### 15.8 Margin allocation (REQ-T16)

```
For each selected demat d:
    cushion_d = max(d.balance × cushion_pct/100, cushion_min)
    deployable_d = max(0, d.balance - cushion_d)

total_deployable = Σ deployable_d
cap = (budget_cr × 1_00_00_000) capped to total_deployable    # 0 = no cap

For each selected demat d:
    allocated_d = cap × deployable_d / total_deployable        # weighted
```

### 15.9 Live margin gauge (REQ-T130)

```
required = Σ legs (margin_per_lot × lots)
free      = total - used_by_active - blocked_by_orders
exceeds   = required > free
```

### 15.10 P&L

```
For each leg:
    signed = +1 if side==SELL else -1
    realised_per_unit = signed × (entry_price - exit_price)
    realised_leg = realised_per_unit × units

strategy_realised = Σ realised_leg - brokerage - STT - fees - GST - stamp
```

---

## 16. SEBI / NSE / BSE compliance — non-negotiable

**REQ-C01.** Algo-ID tag on every order. Reject submission if missing.

**REQ-C02.** Order rate cap ≤ 8/sec/user, 20/sec global. Token bucket in Redis.

**REQ-C03.** OTR monitored live. Auto-halt user's order submission at OTR ≥ 100.

**REQ-C04.** Freeze qty enforcement: NIFTY 1,800 / SENSEX 1,000. Orders > freeze auto-iceberg.

**REQ-C05.** No MARKET orders on options (regardless of underlying). UI disables, backend rejects.

**REQ-C06.** Audit log is **append-only** with hash chain. PG triggers block UPDATE/DELETE on the audit\_log table. Daily anchor to S3.

**REQ-C07.** Two-person approval for orders ≥ 5 lots (configurable). The first user submits, a second admin approves; only then does the order leave the system.

**REQ-C08.** Pre-trade RMS rejects: over-margin, over-position-limit, missing algo-ID, invalid strike, expired session, IP not in allowlist.

**REQ-C09.** Daily SEBI compliance report (XML) generated at 18:00 IST, archived for 7 years.

**REQ-C10.** All broker tokens encrypted at rest with Fernet. Rotation every 90 days.

---

## 17. Acceptance test list (smoke tests)

The dev should run all of these manually before delivering. Each must pass.

### Auth

- ☐ Login succeeds with valid creds + correct TOTP.
- ☐ Login fails after 5 wrong attempts → lockout message.
- ☐ Logout clears session; revisiting requires re-login.

### Default Strategy CTA

- ☐ Pick NIFTY → CE/PE strikes auto-update to round-away values from current spot.
- ☐ Pick SENSEX → values swap to SENSEX lot/grid/spot.
- ☐ "Load Strategy" → confirm modal with full details → OK → legs section pre-filled, no orders placed.
- ☐ "Load + Execute" → confirm modal → type EXECUTE → orders submitted toast → redirect to /strategy/:id.

### Manual builder

- ☐ Add Leg button works; new leg appears in table.
- ☐ B/S, Type toggles colour the row.
- ☐ Strike +/- respects 50/100 grid by underlying.
- ☐ Lots 1–30 in dropdown.
- ☐ LTP read-only; Trade Price editable.
- ☐ Trigger mode = Combined → Add Leg disables at 2 legs; existing extra legs trim with toast.
- ☐ Trigger mode = Per-leg → Trade Price gets accent border.
- ☐ Trigger mode = Per ₹1Cr → live ratio updates; chip flips to MET when threshold crossed.

### Margin

- ☐ Free margin strip refreshes every 15 s.
- ☐ Live gauge bar turns red when required > free.
- ☐ Start / Execute disabled when over-margin; tooltip explains.

### Multi-broker / SOR

- ☐ Multi-demat toggle reveals checkbox list with cap badges.
- ☐ Multi-broker toggle reveals broker rows with embedded demats.
- ☐ Allocation row appears with single-sentence input shape.
- ☐ Allocation preview updates live as inputs change.

### Exit rules

- ☐ SL / Target / Square-off / MTM-DD always visible.
- ☐ Trailing SL toggle reveals fields.

- ☐ Lock-in profits toggle reveals field.
- ☐ **Spot proximity** toggle reveals Distance type / Threshold / Scope; helper text updates.
- ☐ Dead-man hidden under Advanced.

## Reports

- ☐ Filters apply server-side; URL updates with query params.
- ☐ Live + Paper rows distinguishable.
- ☐ Export CSV downloads filtered rows.
- ☐ Click trade row → navigates to audit entry.

## Dashboard

- ☐ KPI cards update live.
- ☐ Active strategies table sortable.
- ☐ Click a row → /strategy/:id.
- ☐ Today's curve matches sum of strategy P&Ls in /reports for today.

## Permissions

- ☐ Default-only user sees ONLY: margin strip, broker picker, Default CTA, Cancel.
- ☐ Manual builder, Save Draft, Start, Execute Now all hidden for default-only.

## Compliance

- ☐ Submit > 1,800 unit NIFTY order → backend rejects OR auto-icebergs.
- ☐ Submit MARKET order on option → rejected.
- ☐ Submit 6-lot order → 2-person approval modal blocks; 2nd admin approves → fires.

# 18. Build phasing (suggested)

The product is large. Suggested phasing if the dev wants to incrementalise:

## Phase A (4 weeks) — MVP single-broker live

- Auth + 2FA + IP allowlist
- Paper broker + Zerodha live broker
- Default Strategy CTA only (no manual builder)
- Free margin strip + single demat
- Combined  $\Sigma$  trigger only
- Stop Loss + Target + Square-off + MTM-DD exits
- Audit log
- Basic reports (Trades + Strategies tabs)
- Dashboard with the 4 KPI cards

## Phase B (3 weeks) — Manual builder

- Manual leg table + expanded quote
- All 4 trigger modes
- Strike Selector (manual + automatic)
- Entry time window
- Trailing SL + Lock-in + Spot proximity exits

- Multi-demat (single broker)
- Confirm modals everywhere
- Reports: By Demat tab

### Phase C (3 weeks) — Multi-broker + advanced

- Axis + Monarch + JM broker adapters
- Multi-broker SOR + margin allocator
- Per-broker margin reconciliation
- Reports: By Portfolio + Tax tabs
- Notifications: all channels
- Admin: Audit Browser + EOD recon

### Phase D (2 weeks) — Polish + go-live

- PWA install
- Two-person approval flow
- Templates page
- Analytics page
- Mobile breakpoints
- Load testing
- SEBI compliance XML export
- Production deploy + monitoring

---

## 19. Visual reference

The reference UI in this repo ( `frontend/` ) implements all the above. To see it:

```
cd frontend
npm install
npm run dev
# open http://localhost:5173
```

Use the dev login bypass (in `HANDOFF.md` §7.3) if backend is not running.

Pages to walk through:

- `/` (Dashboard)
- `/trade` (the main page — every section is visible here)
- `/strategy/:id` (live monitor when a strategy is running)
- `/reports/*` (5 tabs)
- `/templates` , `/analytics` , `/settings` , `/admin/audit`

Take screenshots of each and use them as the visual spec. The dev does not need to copy this code — they need to **match the UX**. Backend, naming, file structure, framework — all the dev's choice.

---

## 20. Open questions

Tracked in `HANDOFF.md` §11. Repeated here so the dev sees them in context:

1. Default Strategy CTA also on `/strategy/new` , or only `/trade` ?

2. Multi-broker SOR — broker session goes dark mid-execution: cancel & re-route, or pause?
3. Premium trigger "Per ₹1Cr" — margin denominator: current required, or projected at fill?
4. Entry time window — apply to trigger evaluation only, or also force-exit positions outside the window?
5. Margin allocation cushion — released to "free" pool when position closes, or stay reserved for next strategy?

The dev should propose answers (with rationale) in their first design review.

---

## 21. What success looks like

When this product is live, a trader should be able to:

1. Log in, see their free margin, click **Load + Execute** on the Default Strategy CTA, type EXECUTE, and have a NIFTY strangle filled within 5 seconds — with no manual decisions about strikes, lots, or triggers.
2. Open a manual builder, set up a 4-leg iron condor, route it across 3 demats with a ₹15 Cr budget keeping ₹5L free per demat, gate entry to "combined premium  $\geq$  ₹50/Cr margin between 09:30–10:30", and walk away — confident the system respects every rule.
3. End the day at /reports, export a CSV split by live/paper/demat, hand it to their CA, and have nothing missing.
4. Get a WhatsApp alert the moment any strategy hits MTM-DD or spot encroaches on a strike, and tap a deep link that takes them straight to the running monitor.

If the dev's build delivers all four flows above without us ever editing their code, that's success.

---

End of scope. Questions: [rohan@navingroup.in](mailto:rohan@navingroup.in).